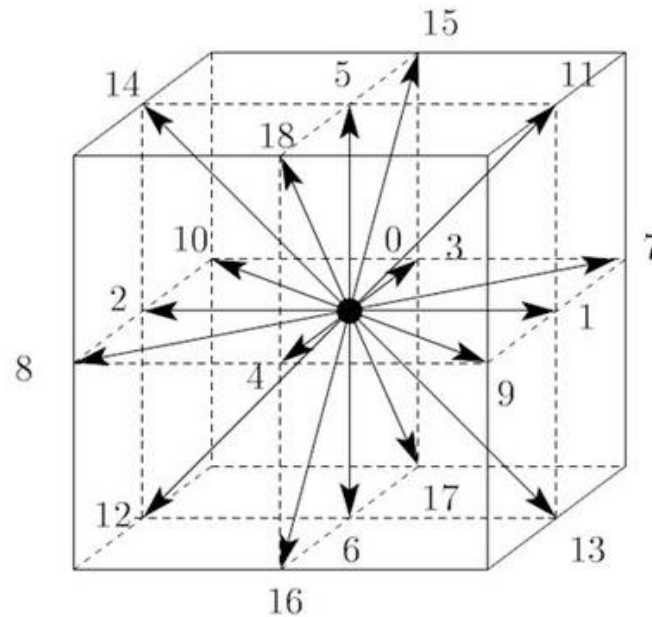




Lattice Boltzmann Method

Dr. Alessandro De Rosis

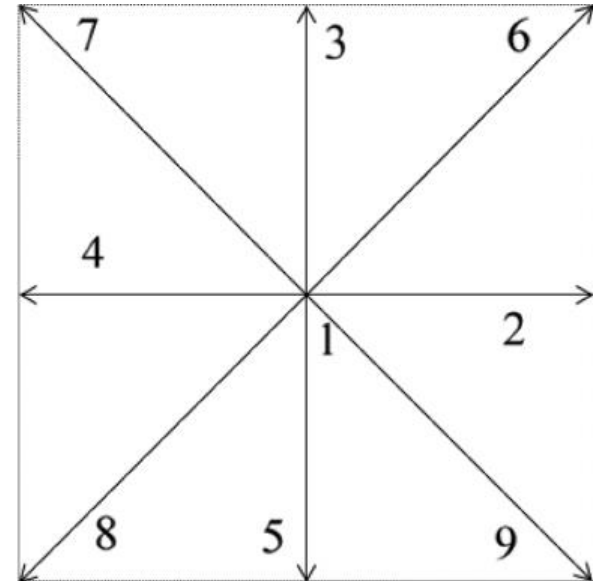
Slides adapted from Dr. Alistair Revell and Dr. Joseph O'Connor

Practical LBM Calculation

- **This is a D2Q9 model**

- **Main user options are:**
 - Spatial resolution
 - Lattice velocity
 - Relaxation time $\tau = 1/\omega$

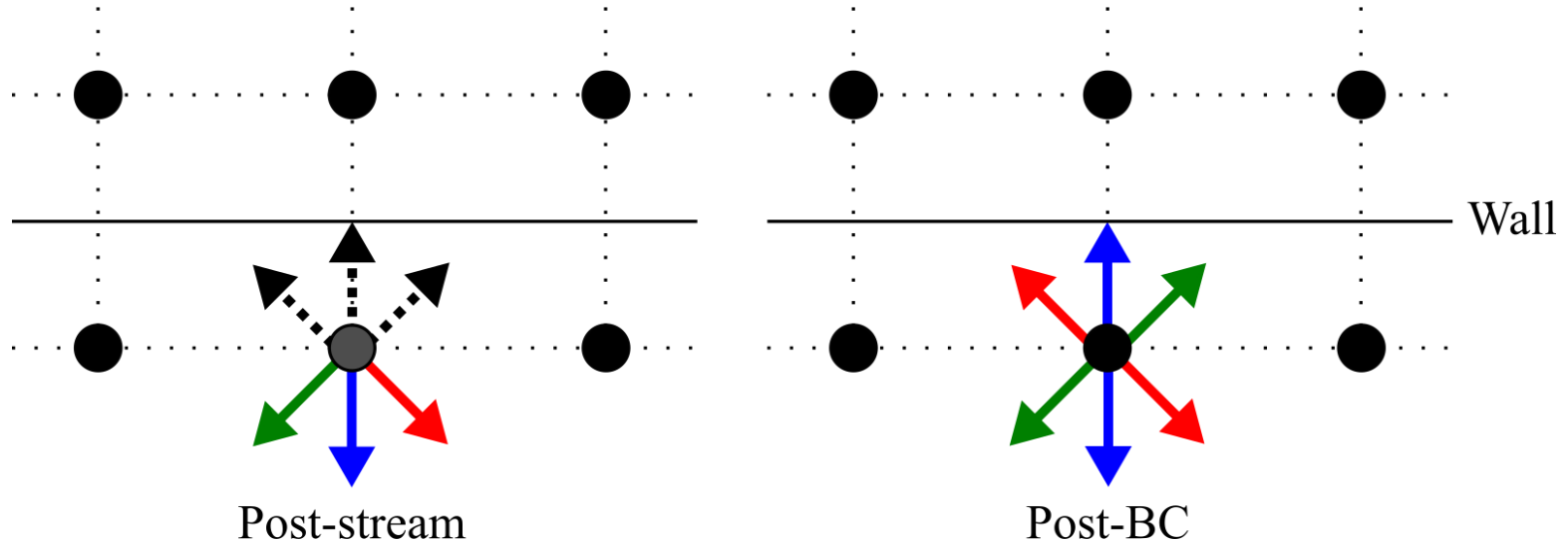
- **Lab requires exploration of basic code**
 - Implementation of basic boundary conditions
 - Testing of various simple flows
 - Opportunity to input more complex geometry



Wall: Bounceback

- **Simple specification of no-slip walls is possible using bounceback**

- Wall is implied 'between nodes'
- Black dashed lines are unknown after streaming step
- Green, red, and blue directions have bounced back off the wall



- **Simple to code (just switch populations)**

- Complex geometry is thus very easy

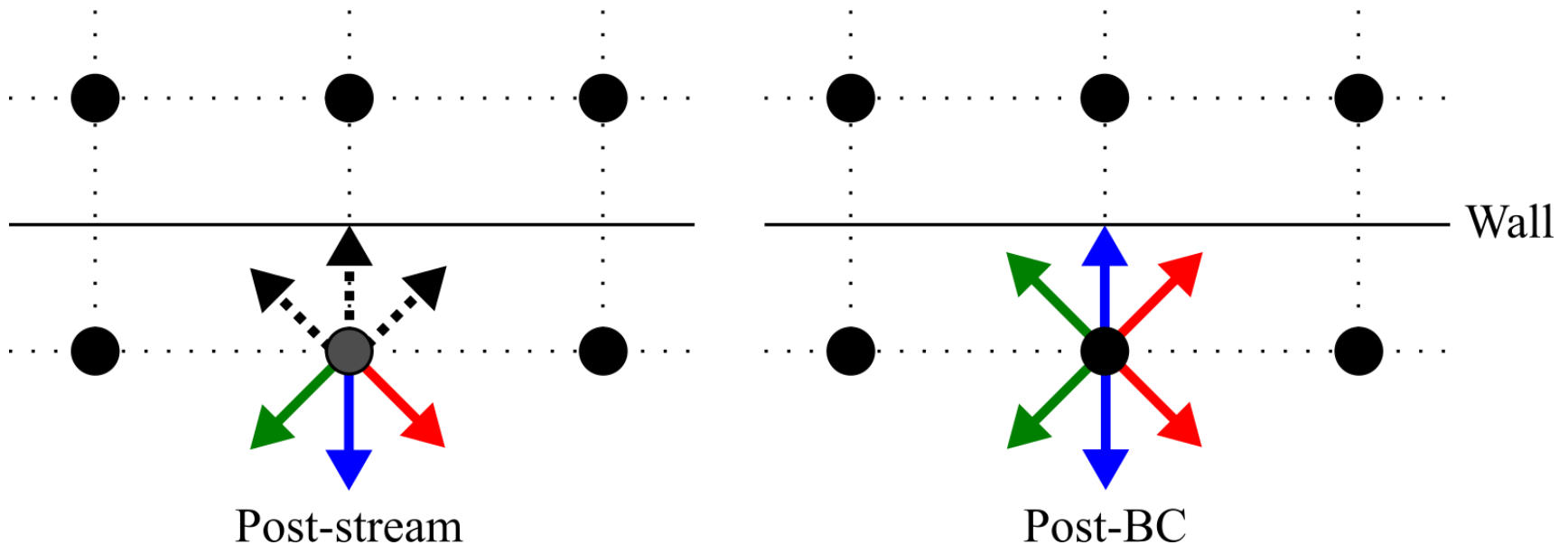
- **However this is only 1st order accurate**

Symmetry

- $\mathbf{u} \cdot \mathbf{n} = 0$ and $du/dn = 0$

- Also called a 'slip wall'

- Populations are mirrored across pseudo boundary



Non-Dimensionality

- It is usually preferable to deal with nondimensional numbers
- For example, the Navier Stokes equations:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \rho \nu \nabla^2 \mathbf{u}$$

- Can be re-written as:

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \tilde{t}} + (\tilde{\mathbf{u}} \cdot \tilde{\nabla}) \tilde{\mathbf{u}} = -\tilde{\nabla} \tilde{p} + \frac{1}{Re} \tilde{\nabla}^2 \tilde{\mathbf{u}}$$

$$\tilde{\mathbf{u}} = \mathbf{u}/u_m, \quad \tilde{p} = p/(\rho u_m^2), \quad \tilde{t} = t u_m/H, \quad \tilde{\nabla} = \nabla H, \quad Re = \frac{u_m H}{\nu}$$

- This can help us interpret the most significant terms
 - For example, at high Reynolds numbers the viscous term becomes negligible

- The more general concept is:

$$Q = \tilde{Q} \times C_Q$$

Where:

Q is the physical value

$\tilde{Q}$ is the dimensionless value

C_Q is the conversion factor

Discretising LBM Problems

- Generally, LBM uses unit discretisation in space and time:

$$\Delta x = \Delta t = 1$$

- This is different to typical CFD, and requires non-dimensionalisation
- The conversions between physical and non-dimensional values are:
 - Remember we use tilde notation to denote the dimensionless (lattice) quantities

$$L = \tilde{L} \times C_L;$$

$$U = \tilde{U} \times C_U;$$

$$\Delta x = 1 \times C_L = L/\tilde{L};$$

$$\Delta t = 1 \times C_T = 1 \times C_L/C_U = L\tilde{U}/\tilde{L}U;$$

$$v = \tilde{v} \times C_v;$$

$$C_v = C_U \cdot C_L = LU/\tilde{L}\tilde{U};$$

$$\tilde{v} = \frac{1}{3} \left(\tau - \frac{1}{2} \right);$$

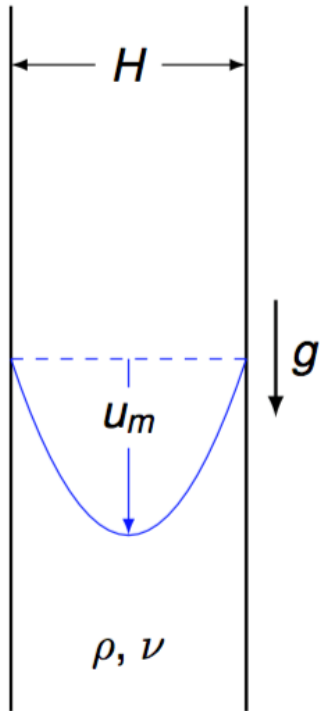
$$v = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \times LU/\tilde{L}\tilde{U};$$

Should be > 0.5

Should be $\ll c_s = 1/\sqrt{3} \rightarrow \text{Ma} \ll 1$

Example

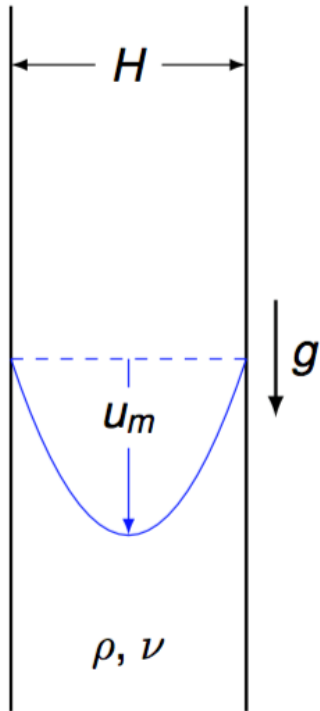
- Consider case of a vertical channel flow driven by gravity



1. Compute physical scales
2. Select spatial resolution and decide τ
3. Find C_L
4. Find corresponding $\tilde{U}$

Example

- Consider case of a vertical channel flow driven by gravity



1. Compute physical scales

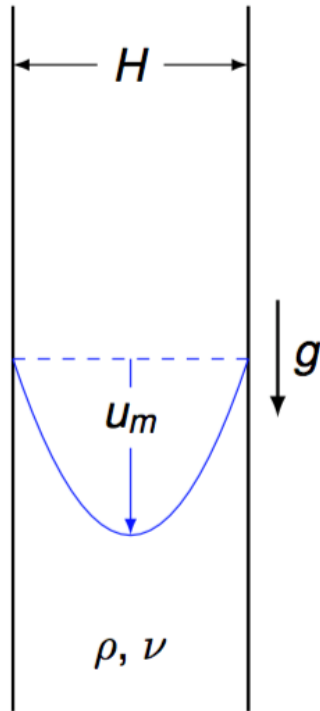
$$\begin{aligned}H &= 10^{-3} \text{ m} \\ \nu &= 10^{-6} \text{ m}^2/\text{s} \\ \rho &= 10^3 \text{ kg/m}^3 \\ g &= 10 \text{ m/s}^2\end{aligned}$$

$$u_m = \frac{gH^2}{8\nu} = 1.25 \text{ m}^2/\text{s}$$

$$Re = 1250$$

Example

- Consider case of a vertical channel flow driven by gravity



- Select spatial resolution and decide τ
- Find C_L
- Find corresponding $\tilde{U}$

$$\tilde{L} = 20, \quad C_L = 5 \times 10^{-5} m, \quad \tau = 0.6$$

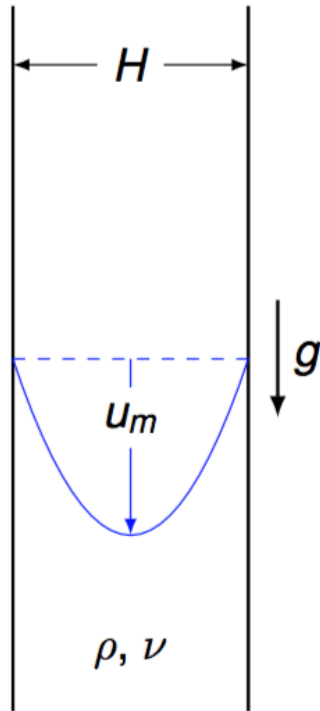
$$v = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \times C_U \cdot C_L = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \times \frac{U}{\tilde{U}} \cdot C_L$$

$$\tilde{U} = 2.083$$

Lattice velocity exceeds speed of sound ($1/\sqrt{3}$)

Example

- Consider case of a vertical channel flow driven by gravity



2. Adjust spatial resolution and τ

$$\tilde{L} = 100, \quad C_L = 1 \times 10^{-5} m, \quad \tau = 0.55$$

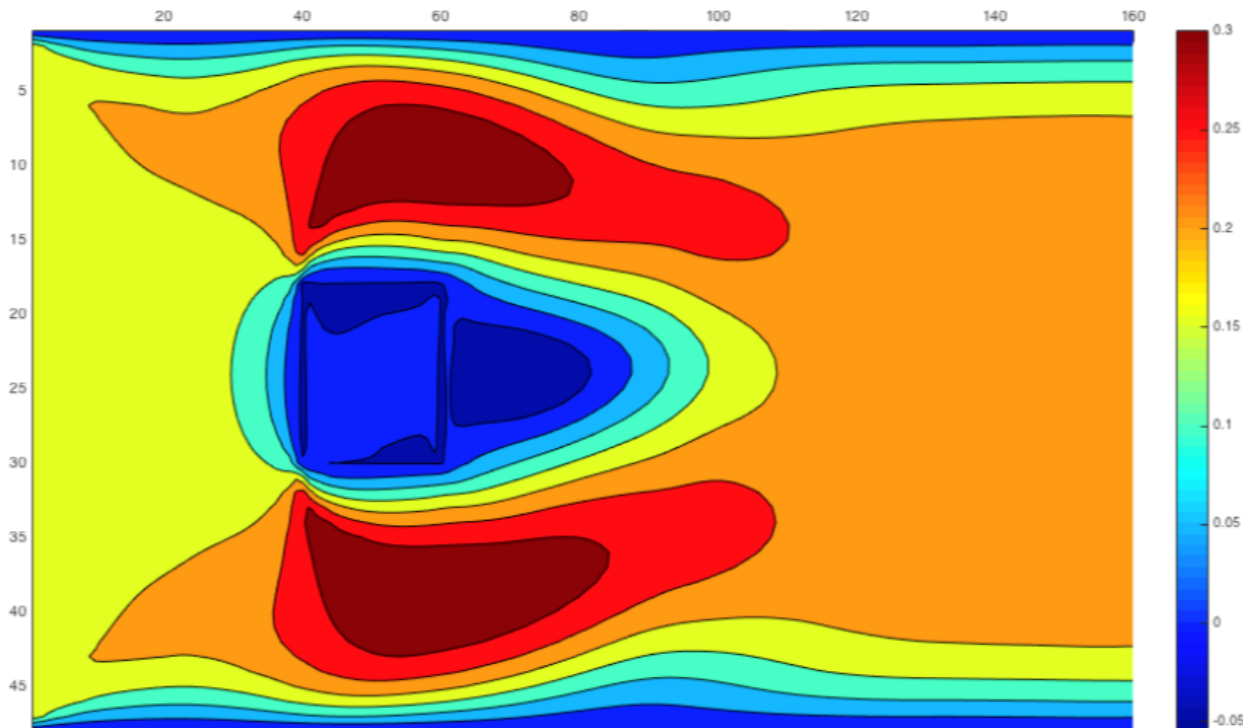
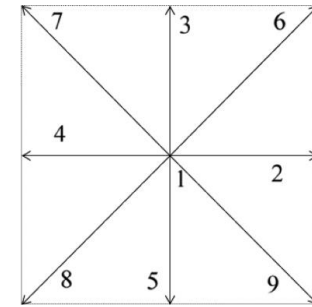
$$v = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \times C_U \cdot C_L = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \times \frac{U}{\tilde{U}} \cdot C_L$$

$$\tilde{U} = 0.2083$$

We can increase the resolution and/or reduce the relaxation time (τ) to decrease the lattice velocity

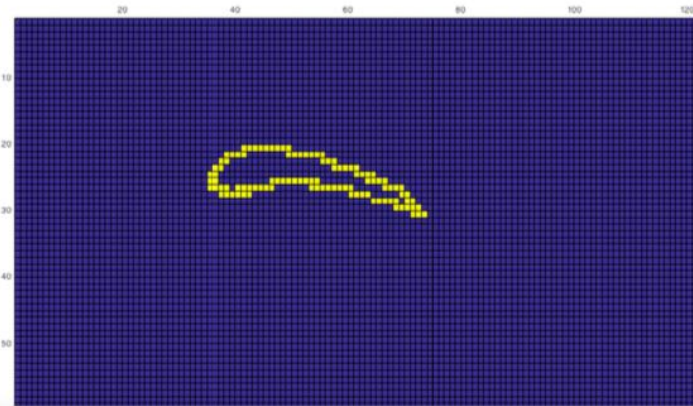
Implementing Bounceback

- Need to refer to velocity model
- Then can either define shapes using code

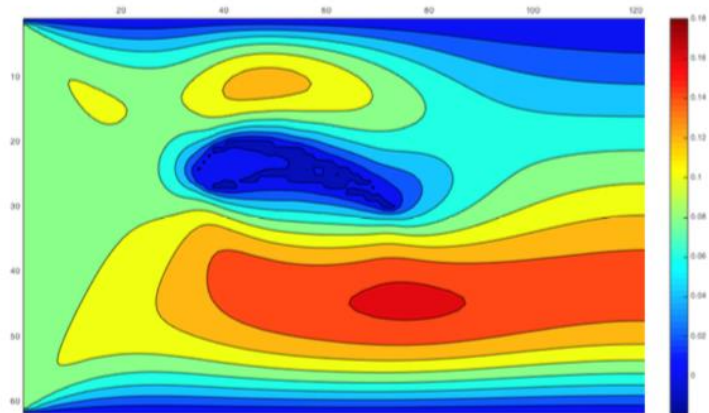


Implementing Bounceback

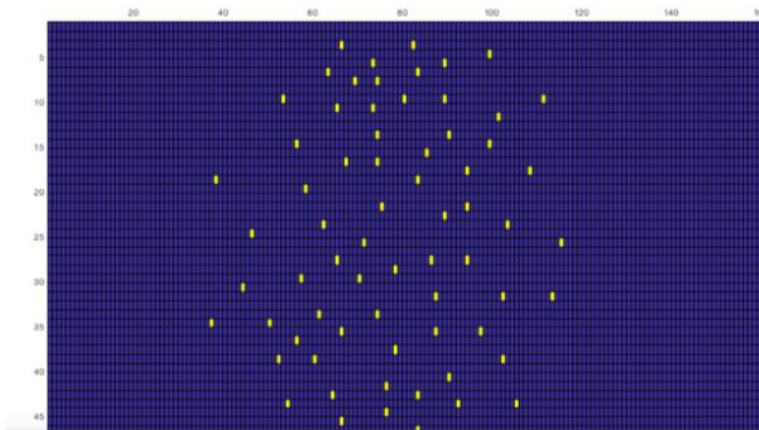
- Or using mouse to input wall locations



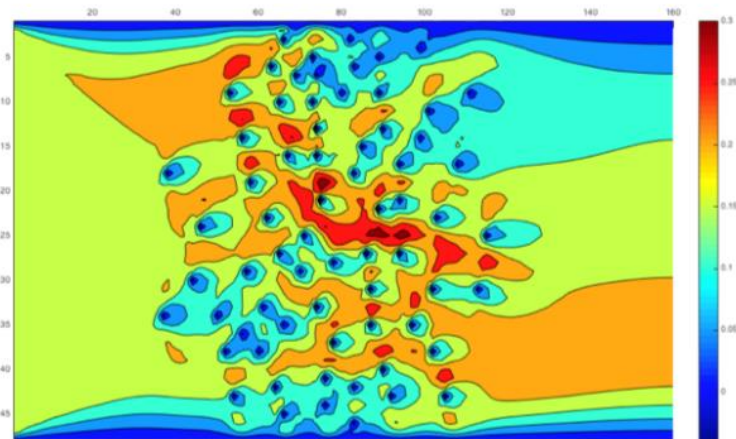
Grid displaying fluid sites (blue) and wall sites (yellow)



Corresponding Flow



Another example



Corresponding Flow